

Programme : Diploma in Cloud Computing and Big Data	
Course Code : 6272D	Course Title: Data Security
Semester : 6	Credits: 4
Course Category: Open Elective Course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To Provide the fundamental concepts of Data Security.
- To Impart the idea of Cryptography and Encryption Algorithms
- To Impart the basics of User Authentication and Access control.
- To Provide concepts of various malicious softwares, intrusion detection systems and firewalls

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge in Computer systems	1008	Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic concepts of Data Security.	12	Understanding
CO2	Explain Cryptography and its application in Data Security	16	Understanding
CO3	Describe the basics of User Authentication and Access control	18	Understanding
CO4	Summarize various malicious softwares, intrusion detection systems and firewalls	12	Understanding

	Series Test	2	
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CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic concepts of Data Security.		
M1.01	Define Data Security	1	Remembering
M1.02	Explain Security Triads	2	Understanding
M1.03	Outline other terminologies related to data security	2	Understanding
M1.04	Describe the Model of Computer Security	3	Understanding
M1.05	Illustrate the OSI Security Architecture	2	Understanding
M1.06	Explain Data breach and how to prevent and mitigate from Data breach	2	Understanding

Contents :

Data Security Definition, Security Triads-Confidentiality, Integrity, Availability
 Basic Terminologies-Authenticity, Accountability, Data privacy and Ethics
 Model of Computer Security- Assets, Threats, Attacks, Risk and Countermeasures
 OSI Security Architecture
 Data breach, How to prevent and mitigate Data breach

CO2	Explain Cryptography and its application in Data Security		
M2.01	Differentiate Cryptology, cryptography and cryptanalysis.	1	Understanding

M2.02	Classify different types of Cryptography	3	Understanding
M2.03	Explain block and Stream Cipher encryption, Substitution ciphers and Transposition ciphers	2	Understanding
M2.04	summarize the strength and application of Symmetric Encryption Algorithms	2	Understanding
M2.05	Outline the applications of Public key cryptography.	3	Understanding
M2.06	Describe Hash functions and list various hash functions	2	Understanding
M2.07	Compare the various message authentication techniques	3	Understanding
	Series Test – I	1	

Contents :

Comparison of Cryptology, cryptography and cryptanalysis

Different types of Cryptography- Symmetric Key Cryptography, Asymmetric Key Cryptography- Simplified models of Symmetric encryption and Asymmetric encryption (public-key cryptography) with block diagrams.

Block and Stream Cipher encryption, Substitution ciphers and Transposition ciphers

Strength and application of Common Symmetric Encryption Algorithms- DES, Triple DES and AES

Applications of Public key cryptography- digital signature, public key certificates and Digital Envelope

Hash Functions- List various hash functions

Message authentication techniques- without encryption(using MAC code), with symmetric encryption, with asymmetric encryption and with hash functions

CO3	Describe the basics of User Authentication and Access control		
M3.01	Illustrate the Model for Electronic User Authentication	2	Understanding
M3.02	Describe the various means of user authentication.	3	Understanding
M3.03	Explain the general forms of attacks against Password based authentication and its countermeasures.	2	Understanding
M3.04	Illustrate use of Hashed Passwords and Password File Access Control	3	Understanding
M3.05	Summarize the accuracy of Biometric based authentication.	2	Understanding
M3.06	Explain the various security issues for User Authentication	2	Understanding

M3.07	Outline Access Control, it's basic elements and policies	4	Understanding
Contents : User Authentication- Definition, Model for Electronic User Authentication Means of user authentication- Password based, Token based and Biometric based Authentication. General forms of attacks against Password based authentication- countermeasures, Hashed Passwords and Password File Access Control Accuracy of Biometric based authentication. Security issues for user Authentication- Client attack, Host attack, Eavesdropping, Replay attack, Denial of service Access Control- Definition, Basic Elements of Access Control - subject, object, and access right List various Access Control Policies- Discretionary access control (DAC), Mandatory access control (MAC), Role-based access control (RBAC), Attribute-based access control (ABAC)			
CO4	Summarize various malicious software, intrusion detection systems and firewalls		
M4.01	Classify different types of Malicious software(Malware).	1	Understanding
M4.02	Describe the precautions to prevent malware attacks	1	Understanding
M4.03	Explain various classes of intruders, and intruder behavior patterns.	1	Understanding
M4.04	Summarize intrusion detection systems	5	Understanding
M4.05	Outline about Firewall	4	Understanding
	Series Test – II	1	
Contents : Different types of Malicious softwares- Virus- Trojans- Spyware- Adware- Rootkits- Ransomware- Worms-Keyloggers Precautions to prevent malware attacks Various classes of intruders- based on motivations and based on skill levels, Intruder behavior patterns, working of Intrusion Detection System, Classification based on source and type- Host based, Network based and Networked IDS, Classification based on Analysis Approaches-Anomaly based, Signature based IDS, Firewall - Definition, Need for firewall- Various characteristics of a Firewall, Various types of Firewalls- Packet filtering, Stateful Inspection, Application Level Gateway and Circuit Level Gateway			

Text / Reference

T/R	Book Title/Author
T1	Computer Security- Principles and Practice - Author: William Stallings & Lawrie Brown Publisher: Pearson Prentice Hall 2010
R1	Cryptography and Security - Author: C K Chyamala, N Harini & Dr T R Padmanabhan Publisher: Wiley – India 2010
R2	Network Security –M.V. Arun Kumar, USP2011 First Edition

Online Resources

Sl.No	Website Link
1	https://www.titanfile.com/blog/types-of-computer-malware/
2	https://www.techtarget.com/searchsecurity/definition/malware